ST5202 —Applied Regression Analysis
Lecture Notes 8: Remedial Measures

Chen Zehua and Wang Wanjie

Department of Statistics and Data Science

March 22, 2026

Contents

- Overview of remedial measures
- Nonlinearity
- Heteroscedasticity
- Multicollinearity
- Non-normality

Remedy for nonlinearity

- If diagnostic plots show systematic curvature, revisit the functional form of predictors.
- In Lecture 5, we have introduced some common remedies for nonlinearity, depending on the plot pattern:
  - adding polynomial terms such as x^2 or x^3,
  - transforming predictors such as $\log x$, $1/x$, or $\sqrt{x}$,
  - adding interaction terms when the trend depends on another predictor,
  - using piecewise linear terms when the slope changes across regions.
- After modifying the model, refit it and check the residual plots again.

LRM with unequal variances

- When σ_i^2 depends on i, an unequal variance model can be considered.
- An unequal variance model is given by

$$\boldsymbol{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon},$$

where $\boldsymbol{\epsilon}$ has the variance matrix

$$\Sigma = \sigma^2 \begin{pmatrix} w_1^{-1} & 0 & \cdots & 0 \\ 0 & w_2^{-1} & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & w_n^{-1} \end{pmatrix},$$

where $w_i = \sigma^2/\sigma_i^2$ are unequal weights.

When the weights are known

When the w_i's are known, the unequal variance model can be transformed into an equal variance model.

- Define the weight matrix (predictors with a larger variance has a smaller weight to eliminate the effect)

$$W = \begin{pmatrix} w_1 & 0 & \cdots & 0 \\ 0 & w_2 & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & w_n \end{pmatrix}.$$

- Multiply $W^{1/2}$ to both sides of $\boldsymbol{y} = X\boldsymbol{\beta} + \boldsymbol{\epsilon}$,

$$W^{1/2}\boldsymbol{y} = W^{1/2}X\boldsymbol{\beta} + W^{1/2}\boldsymbol{\epsilon}.$$

Let $\tilde{\boldsymbol{y}} = W^{1/2}\boldsymbol{y}$, $\tilde{X} = W^{1/2}X$ and $\tilde{\boldsymbol{\epsilon}} = W^{1/2}\boldsymbol{\epsilon}$. Then $\mathrm{Var}(\tilde{\boldsymbol{\epsilon}}) = W^{1/2}\Sigma W^{1/2} = \sigma^2 I$. and the model $\tilde{\boldsymbol{y}} = \tilde{X}\boldsymbol{\beta} + \tilde{\boldsymbol{\epsilon}}$ is a constant variance model.

Weighted least square estimates (WLSE)

- Minimizing $\|\tilde{\boldsymbol{y}} - \tilde{X}\boldsymbol{\beta}\|_2^2$, we obtain the estimate of $\boldsymbol{\beta}$ as

$$\hat{\boldsymbol{\beta}}_w = (\tilde{X}^\top \tilde{X})^{-1}\tilde{X}^\top \tilde{\boldsymbol{y}} = (X^\top W X)^{-1} X^\top W \boldsymbol{y}.$$

  The estimate is called the weighted LSE (WLSE).

- The target function $\|\tilde{\boldsymbol{y}} - \tilde{X}\boldsymbol{\beta}\|_2^2$ can be expressed explicitly as

$$\|\tilde{\boldsymbol{y}} - \tilde{X}\boldsymbol{\beta}\|_2^2 = \sum_{i=1}^{n} w_i (y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta})^2.$$

  The weight w_i reflects the relative importance of $(y_i - \boldsymbol{x}_i^\top \boldsymbol{\beta})^2$ in the estimation. The larger the variance of the ith term, the smaller the corresponding weight, and vise versa.

The distribution of WLSE

- For the WLSE $\hat{\boldsymbol{\beta}}_w$,

$$E\hat{\boldsymbol{\beta}}_w = \boldsymbol{\beta},\ \mathrm{Var}(\hat{\boldsymbol{\beta}}_w) = \sigma^2(X^\top WX)^{-1}.$$

- The distribution:

$$\hat{\boldsymbol{\beta}}_w \sim N\left(\boldsymbol{\beta}, \sigma^2(X^\top WX)^{-1}\right).$$

- σ^2 is estimated by

$$\hat{\sigma}^2 = \frac{1}{n-p-1}\|\tilde{\boldsymbol{y}} - \tilde{X}\hat{\boldsymbol{\beta}}_w\|_2^2 = \frac{1}{n-p-1}\sum_{i=1}^{n} w_i(y_i - \boldsymbol{x}_i^\top \hat{\boldsymbol{\beta}}_w)^2.$$

- The inference on $\boldsymbol{\beta}$ is made in the same way as that in constant variance LRM.

When the weights are unknown

When the variances are un-equal, the variance is usually a function of the mean. In this case, the weights can be estimated by the following procedure:

1. Fit the regression model by unweighted least squares and obtain the residuals $\boldsymbol{r}$ and the fitted values $\hat{\boldsymbol{y}}$.
2. Regress the log of the absolute residual on the log of fitted values, e.g.,
$$\log|r_i| = \alpha_0 + \alpha_1 \log \hat{y}_i + e_i,$$
if smallest $\hat{y}_i < 0$, replace $\hat{y}_i$ by $\hat{y}_i - \min \hat{y}_i + c$, for some positive constant c.
3. The weights are estimated as $w_i = \hat{y}_i^{-2\hat{\alpha}_1}$ (or $e^{-2\hat{\alpha}_0}\hat{y}_i^{-2\hat{\alpha}_1}$). Note that the constant $e^{-2\hat{\alpha}_0}$ can be ignored.

Unknown weights with replicates

If there are replicates of predictor values (e.g. designed experiment case), the sample variance can be used to estimate w_i:

- The naive method is directly taking the inverse of the sample variance as the weight.
- It is usually possible to estimate the variance as a function of the mean by using the regression model

$$\log s_i = \alpha_0 + \alpha_1 \log |\bar{y}_i| + e_i,$$

where s_i is the sample standard deviation and $\bar{y}_i$ is the sample mean. The weight for the ith predictor values is estimated as $w_i = |\bar{y}_i|^{-2\hat{\alpha}_1}$

Other approaches

Weight estimation in WLS:

- If replicated observations are available, estimate group variances directly and use $w_i \propto 1/s_i^2$.
- Fit a variance function from the data, $\sigma_i^2 = a + b\mu_i^2$, etc. Then use estimated inverse-variance weights.
- Use an iterative procedure: fit the model, estimate the variance pattern from residuals, update weights, and refit.
- Nonparametric estimation: fit $r_i^2 \sim g(y_i)$ using a smooth curve, less interpretable and more tuning-dependent.

Approaches beyond WLS:

- GLS: a generalization of WLS for arbitrary $\Sigma = \mathrm{Var}(\boldsymbol{\epsilon})$.
- Robust standard errors: keep OLS estimates, but adjust standard errors and tests for heteroscedasticity.

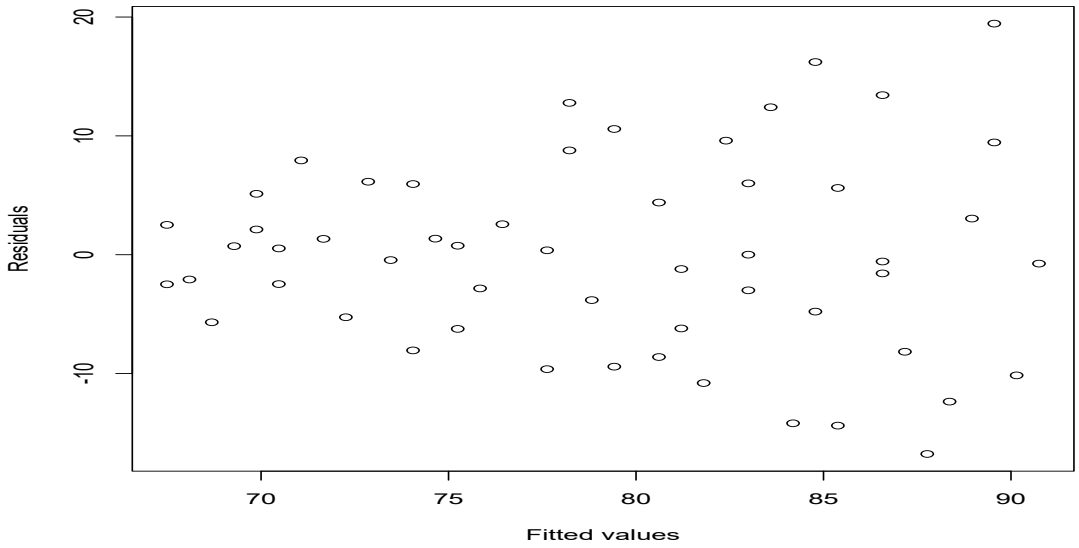
Women's Blood pressure (WBP) example

The example concerns a health study for the relationship between diastolic blood pressure and age among healthy adult women 10 to 60 years old.
The data contains the age and blood pressure of 54 women. The data set is provided in the file `BloodPressure.txt` on Canvas.
We do the following:

- Fit an unweighted regression line and analyze the residual to identify non-constant variance.
- Use the residual to estimate the weights.
- Fit an weighted regression line using the estimated weights.
- Compare the unweighted and weighted regression models.

WBP example: unweighted model and diagnostics

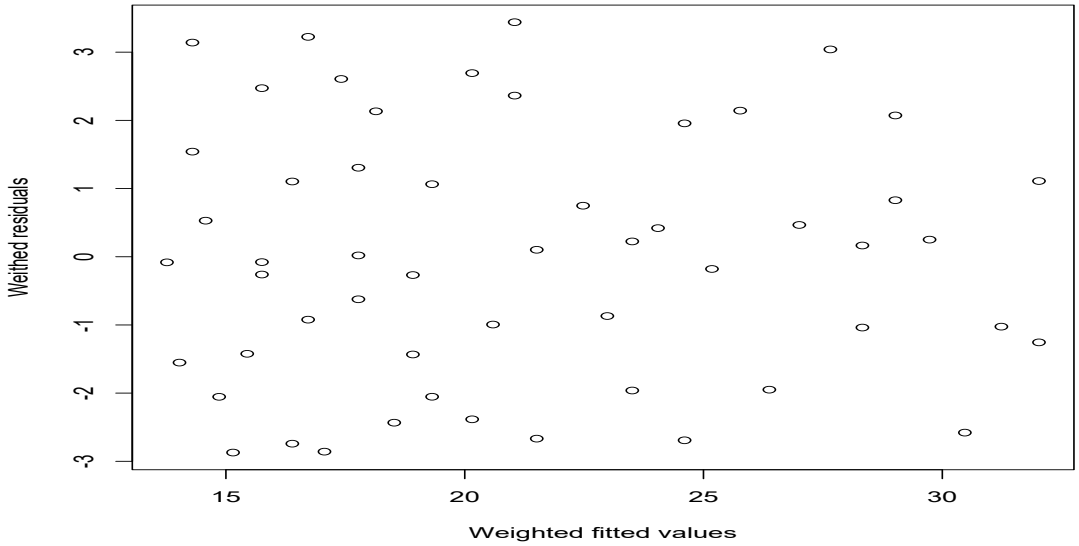
```
WBP.dat = read.table("D://Rsession/WomenBP.txt",header=TRUE)
attach(WBP.dat);y=Pressure; x=Age.women
uw.fit = lm(y~x);r = uw.fit$resid;y.h = uw.fit$fitted
plot(y.h,r,xlab="Fitted values",ylab="Residuals")
```



The plot shows that the variance increases as the fitted value increases. The variance is not constant.

WBP example: weighted model and diagnostics

```
wt.fit=lm(log(abs(r))~log(y.h));
s=exp(wt.fit$fitted); w=1/s^2
w.fit=lm(y~x,weight=w); r=w.fit$resid; y.h = w.fit$fitted
r.w=r*sqrt(w); y.w =y.h*sqrt(w)
plot(y.w,r.w,xlab="Weighted fitted values",ylab="Weighted residuals")
```



The transformed model does not show evidence of non-constancy of variance.

A note on the diagnostics of weighted model

- The diagnostics for the WLSE is the diagnostics on the LSE of the transformed model:

$$\tilde{\boldsymbol{y}}(= W^{1/2}\boldsymbol{y}) = W^{1/2}X\boldsymbol{\beta} + W^{1/2}\boldsymbol{\epsilon}.$$

- Using R for WLSE, the estimate parameter is $\hat{\boldsymbol{\beta}}_w$ for $\boldsymbol{\beta}$, the fitted fitted values are $\hat{\boldsymbol{y}}$ for $\boldsymbol{y}$ (not for $\tilde{\boldsymbol{y}}$), and the residuals $\boldsymbol{e} = \hat{\boldsymbol{y}} - X\hat{\boldsymbol{\beta}}_w$ for $\boldsymbol{\epsilon}$.
- We should transform these values to the transformed model:

$$\hat{\boldsymbol{y}}_w = W^{1/2}\hat{\boldsymbol{y}}; \qquad \boldsymbol{e}_w = W^{1/2}\boldsymbol{e}.$$

- The diagnostics for the weighted LSE check constancy of variance refers to that of the e_w's.

WBP Example: Fitted Results

- Results of unweighted models:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 56.15693    3.99367  14.061  < 2e-16
x            0.58003    0.09695   5.983 2.05e-07
Multiple R-squared:  0.4077.
```

- Results of weighted models:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 55.91146    2.82682  19.779  < 2e-16 ***
x            0.58705    0.08361   7.022 4.57e-09 ***
Multiple R-squared:  0.4867.
```

- The weighted estimates usually have smaller standard errors.
- R^2 for the weighted model is the proportion of weighted total sum of squares explained by $(\sqrt{\boldsymbol{w}}, \sqrt{\boldsymbol{w}}\boldsymbol{x})$. The comparison of weighted and un-weighted models by using R^2 does not make much sense.

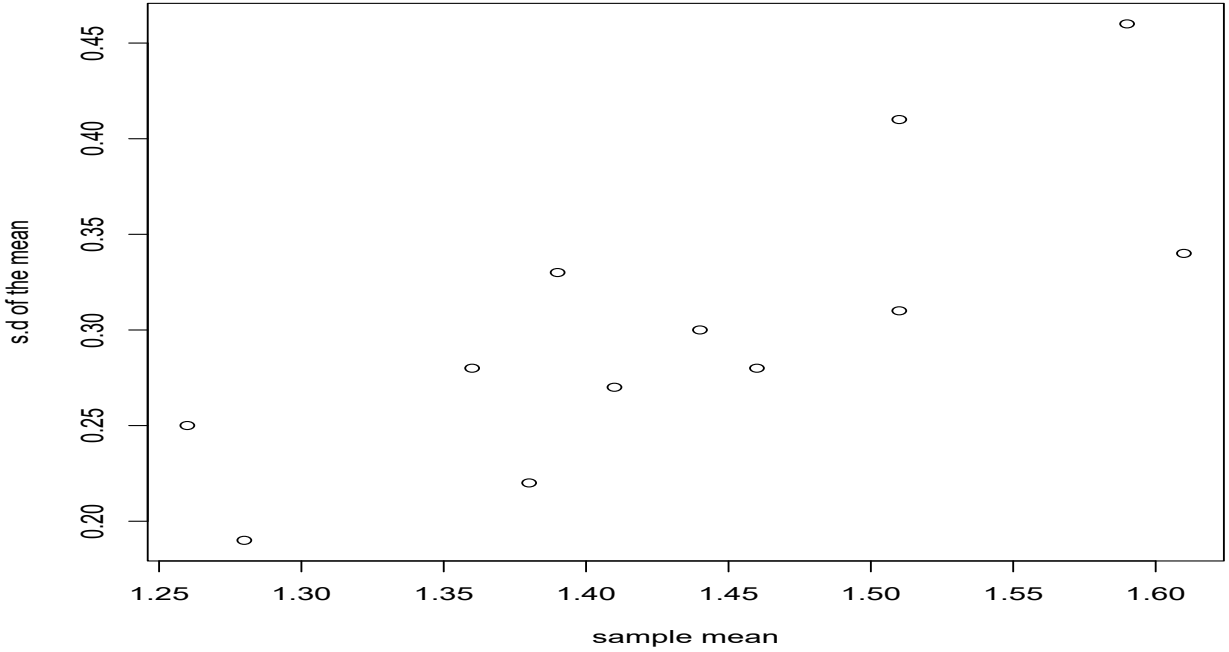
Trial for children with neurological problems (TCNP)

In a study comparing two treatments (T) for children with moderate neurological problems, the response is a measure of neurological functioning obtained eight weeks after the start of the treatment. The children were stratified according to their gender ($G \in \{F, M\}$) and their parent's social class ($S \in \{L, M, H\}$).

The summary data is in the following table:

	Social class	Gender	Treatment 1			Treatment 2		
			n	$\bar{y}$	s	n	$\bar{y}$	s
1	L	F	41	1.38	0.22	40	1.36	0.28
2	L	M	41	1.26	0.25	38	1.28	0.19
3	M	F	33	1.51	0.31	35	1.41	0.27
4	M	M	45	1.46	0.28	46	1.39	0.33
5	H	F	18	1.61	0.34	20	1.51	0.41
6	H	M	23	1.59	0.46	23	1.44	0.30

TCNP Example: s vs. $\bar{y}$



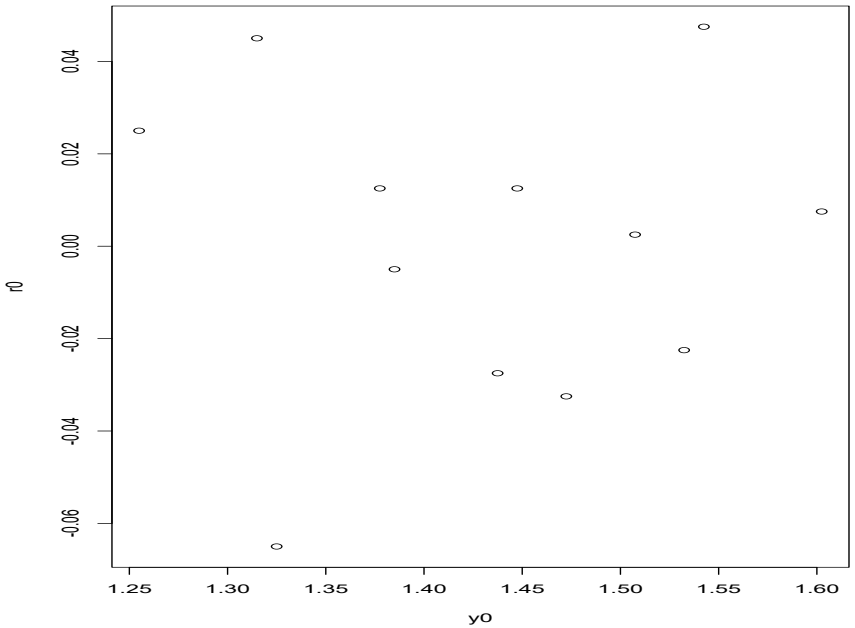
The plot shows that the 12 populations do not have the same variance.

TCNP Example: Preliminary thought

- Consider the regression model with $\bar{y}_i$'s as the response variable.
- The individual observation involved in $\bar{y}_i$ are i.i.d. from a population with mean μ_i and variance σ_i^2. The variance of $\bar{y}_i$ is given by σ_i^2/n_i.
- Two types of weights can be considered:
  (i) If σ_i's do not have any relationship with μ_i, the weights can be taken as $w_i = n_i/s_i^2$.
  (ii) If there is a relationship $\sigma_i = \psi(\mu_i)$, the weights can be taken as $w_i = n_i/\hat{\psi}(\bar{y}_i)$.

TCNP Example: unweighted fitting

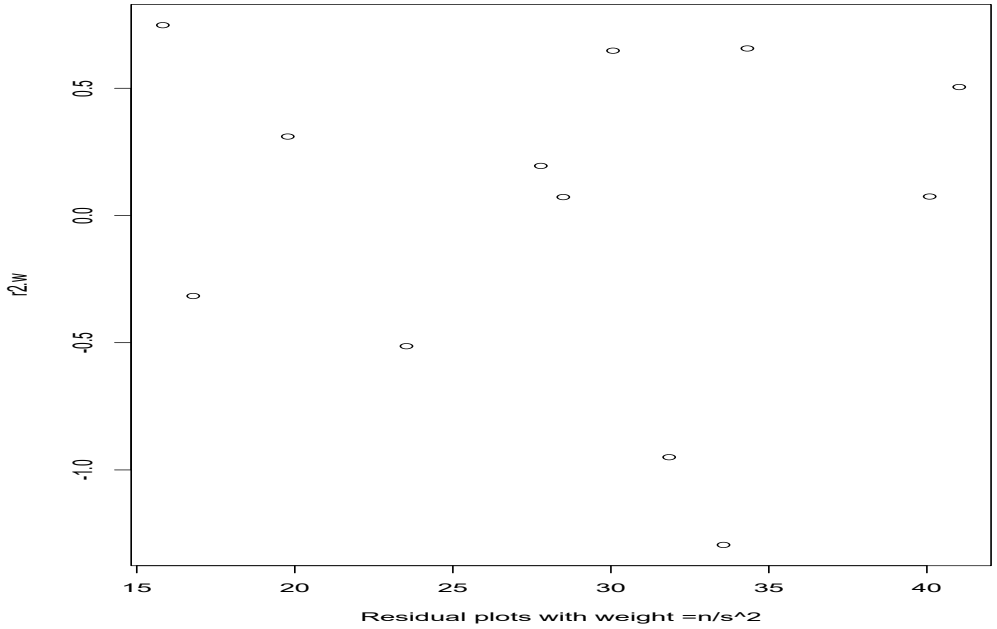
```
fit1=lm(y~T+G+S) #unweighted
r0=fit1$resid; y0=fit1$fitted
plot(y0,r0)
```



The residual plot shows non-constancy of variance.

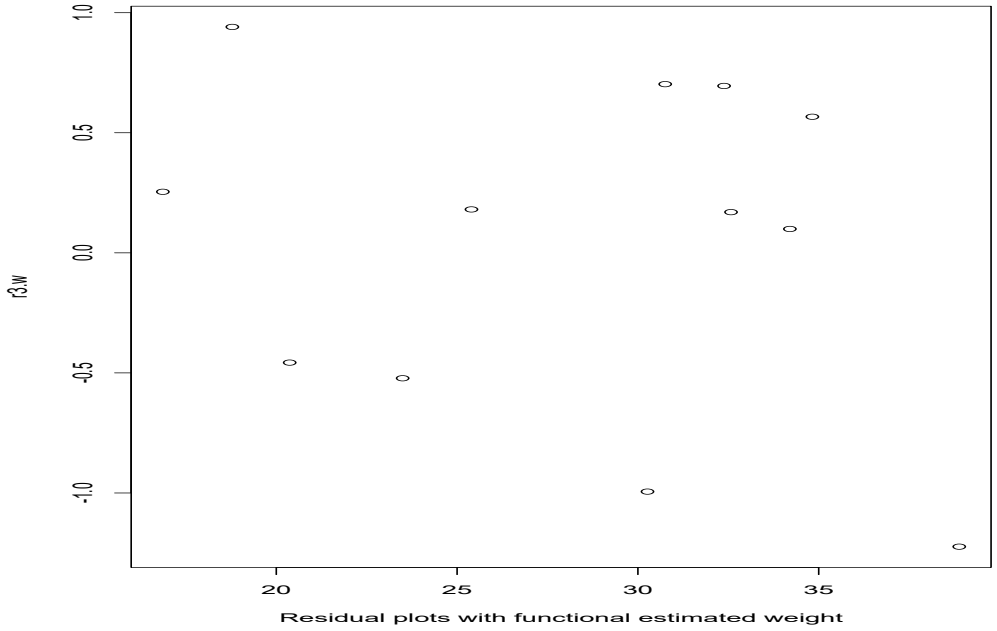
TCNP Example: weight taken as n/s^2.

```
w2=n/s^2
fit2=lm(y~T+G+S,weight=w2)
r2=fit2$resid; y2=fit2$fitted
r2.w=r2*sqrt(w2); y2.w =y2*sqrt(w2)
plot(y2.w,r2.w,xlab="Residual plots with weight =n/s^2")
```



TCNP Example: weight taken as $n/\psi(\bar{y})$.

```
w.fit = lm(log(s)~log(y));log.s =w.fit$fitted; s3 =exp(log.s)
w3=n/s3^2   #group indivicual variance as function of mean
fit3=lm(y~T+G+S,weight=w3);r3=fit3$resid; y3=fit3$fitted
r3.w=r3*sqrt(w3); y3.w =y3*sqrt(w3)
plot(y3.w,r3.w,xlab="Residual plots with functional estimated weight")
```



TCNP Example: results using the two weights

- Both n/s^2 (weight 1)and $n/\psi(\bar{y})$ (weight 2) seem to provide appropriate weights judged by the residual plots.
- The estimated standard deviation of the estimated parameters are given below:

```
            weight 1 weight 2
(Intercept) 0.000476 0.000621
T2          0.000500 0.000580
G2          0.000504 0.000591
S2          0.000595 0.000696
S3          0.001287 0.001365
```

- The estimates with weight n/s^2 have smaller standard deviations and should be preferred.

Remark on choice of weights

- A set of weights is considered appropriate when the weighted residual plot does not show any pattern (i.e., it appears like a random scatter plot).
- The R^2 cannot be used as a criterion for the selection of different weights, since the R^2 with different weights measures different things.
- The standard deviations of the estimated parameters can be used as a reference for the choice among different weights. The one with smaller standard deviations is preferred.
- The cross-validation is the ultimate criterion for the choice of different weights.

Employee salary (ES) example

- In order to establish salary ranges for technical positions, a group of high-tech companies gathered the following data on their employee's salaries
  - y: current salary
  - degree obtained (1 = bechelor, 2 = master, 3 = PhD); years of experience since last degree (x_1); number of persons currently supervised (x_2).
- The data is provided in the file `Salary.txt` on Canvas.
- A preliminary analysis is carried out for model diagnostics. The R-codes for the preliminary analysis are given on the next slide.

ES example: R code for preliminary analysis

```
salary.dat=read.table("D://Rsession/Salary.txt",header=TRUE)
attach(salary.dat)
y=salary;D=factor(degree);x1=YOE;x2=NSP
wk.dat=data.frame(y,D,x1,x2)
uw.fit=glm(y~D+x1+x2,data=wk.dat)
r = uw.fit$resid;yhat=uw.fit$fitted;
par(mfrow=c(2,2))
plot(yhat,r);plot(degree,r);plot(x1,r); plot(x2,r)
hat = hatvalues(uw.fit,type="diagonal")
   infl = influence(uw.fit, do.coef = FALSE)
rsta = rstandard(uw.fit, infl, type = "pearson")
rstu = rstudent(uw.fit, infl, type = "pearson")
cook.d =cooks.distance(uw.fit, infl,res = infl$pear.res,
   dispersion = summary(uw.fit)$dispersion,hat = infl$hat)
par(mfrow=c(2,2))
qqnorm(hat);qqnorm(rstu);qqnorm(cook.d)
n=length(cook.d); cook.d[order(cook.d)][(n-3):n]
rstu[order(rstu)][c(1:3,(n-3):n)]
```

ES example: weighted LSE procedure

- The preliminary analysis reveals that the variances are not constant and that the observations 23 and 43 are influential outliers. In the further analysis, these two points are removed.
- R codes:

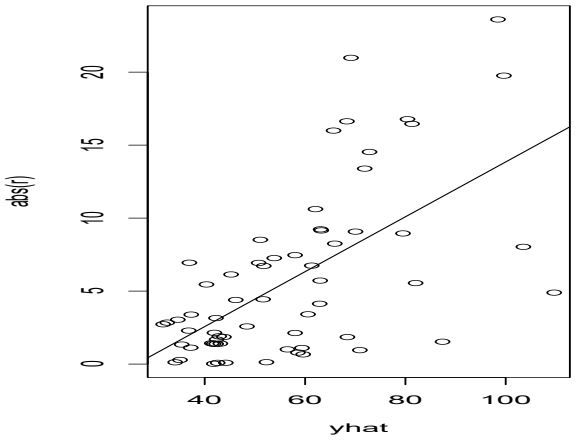
```
wk.dat3 = wk.dat[-c(23,43),]
uw.fit=glm(y~D+x1+x2,data=wk.dat3)
r = uw.fit$resid; yhat=uw.fit$fitted
weight =lm(log(abs(r))~log(yhat)); s=exp(weight$fitted); w=1/s^2
w.fit=lm(y~D+x1+x2,weight=w,data=wk.dat3)
r.w=w.fit$resid*sqrt(w); y.w=w.fit$fitted*sqrt(w)
t.w=lm(abs(r.w)~y.w);t.uw=lm(abs(r)~yhat)
par(mfrow=c(1,2))
plot(yhat, abs(r))
abline(t.uw)
plot(y.w,abs(r.w))
abline(t.w)
```

ES example: Results of weighted LSE

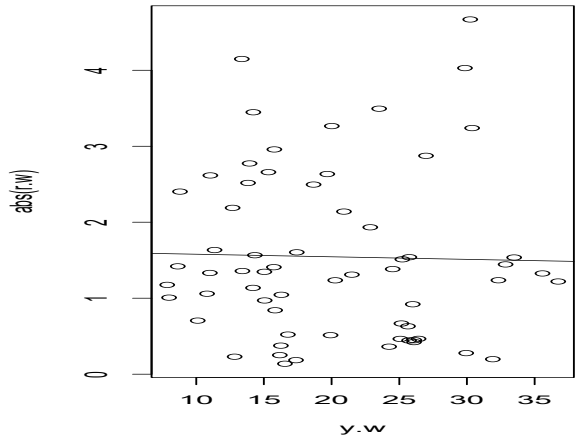
- Results of weighted LSE:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  28.9863     0.8365  34.652  < 2e-16 ***
D2           11.0468     1.0423  10.598 3.45e-15 ***
D3           26.8896     2.2376  12.017  < 2e-16 ***
x1            1.5522     0.1798   8.634 5.36e-12 ***
x2            1.5244     0.4343   3.510 0.000874 ***
```

- Checking for constant variance after weighting.



Unweighted



Weighted

Multicollinearity and its effects

- Multicollinearity refers to the existence of strong correlation among predictor variables.
  Mathematically, there exists j, so that

  $$x_j = a_0\mathbf{1} + a_1x_1 + a_2x_2 + \cdots + a_{j-1}x_{j-1} + a_{j+1}x_{j+1} + \cdots a_px_p.$$

  When the equality holds, we call it a *perfect multicollinearity*.
- Serious multicollinearity causes problems in LSE
  - LSE and Var(LSE) has $(X^\top X)^{-1}$.
  - Serious multicollinearity causes $X^\top X$ to be nearly singular (smallest eigenvalue close to 0 while largest eigenvalue is large), hence the LSE is unstable
  - Serious multicollinearity greatly increases the variance of the LSE

Informal diagnostics for multicollinearity

The following are indications of multicollinearity:

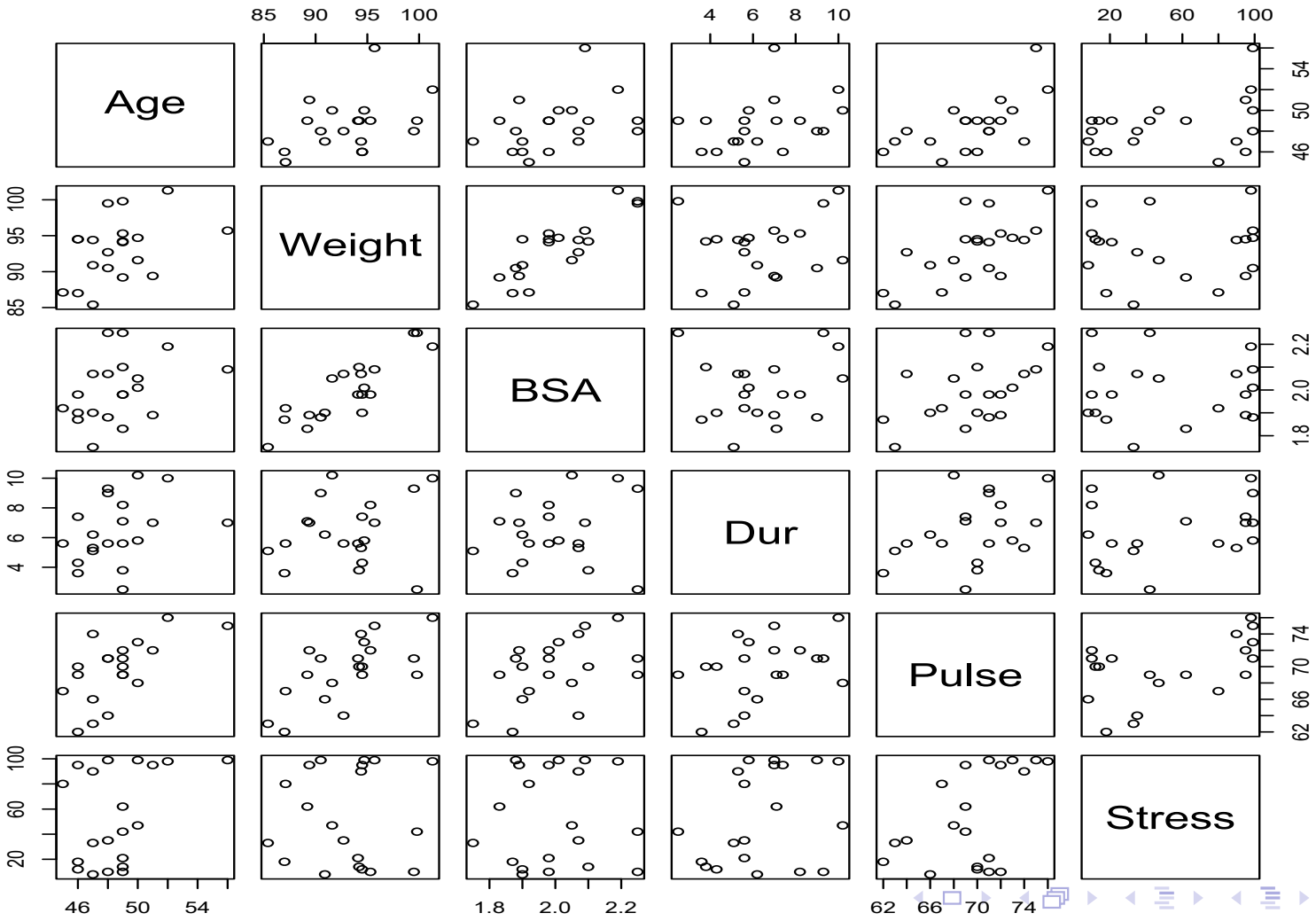
- Large changes in the estimated regression coefficients when a predictor is added or deleted, or an observation is altered or deleted.
- Nonsignificant results in individual test on the regression coefficients for important predictor variables.
- Estimated regression coefficients with an algebraic sign that is opposite of that expected from theoretical consideration or prior experience.
- Large coefficients of sample correlation between pairs of predictors in the correlation matrix.

Another blood pressure (BP) example

- A blood pressure data provided in the file `bloodpressure.txt` on Canvas consists of the following variables: (y = BP, in mm Hg), age (x_1=Age, in years), weight (x_2=Weight, in kg) ,body surface area (x_3=BSA, in sq m) duration of hypertension (x_4=Dur, in years) basal pulse (x_5=Pulse, in beats per minute) stress index (x_6=Stress).
- Correlation matrix of the predictors:

	Age	Weight	BSA	Dur	Pulse	Stress
Age	1.000	0.407	0.378	0.344	0.619	0.368
Weight	0.407	1.000	0.875	0.201	0.659	0.034
BSA	0.378	0.875	1.000	0.131	0.465	0.018
Dur	0.344	0.201	0.131	1.000	0.402	0.312
Pulse	0.619	0.659	0.465	0.402	1.000	0.506
Stress	0.368	0.034	0.018	0.312	0.506	1.000

Scatter plot matrix of the predictors



Coefficient instability after deleting predictors

The results of the fitted full model:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -12.870476   2.556650  -5.034 0.000229 ***
Age           0.703259   0.049606  14.177 2.76e-09 ***
Weight        0.969920   0.063108  15.369 1.02e-09 ***
BSA           3.776491   1.580151   2.390 0.032694 *
Dur           0.068383   0.048441   1.412 0.181534
Pulse        -0.084485   0.051609  -1.637 0.125594
Stress        0.005572   0.003412   1.633 0.126491
```

The results of the model deleting Age:

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept) 4.742003   8.736001   0.543  0.59580
Weight      0.805284   0.242514   3.321  0.00505 **
BSA         9.195931   5.994205   1.534  0.14728
Dur         0.157449   0.187786   0.838  0.41587
Pulse       0.209245   0.184796   1.132  0.27653
Stress      0.006463   0.013338   0.485  0.63551
```

The results of the model deleting Pulse:

```
              Estimate Std. Error t value Pr(>|t|)
(Intercept) -12.587398   2.699492  -4.663 0.000366 ***
Age           0.670659   0.048081  13.948 1.33e-09 ***
Weight        0.899467   0.048847  18.414 3.29e-11 ***
BSA           4.887143   1.510274   3.236 0.005978 **
Dur           0.057500   0.050780   1.132 0.276520
Stress        0.002396   0.002971   0.806 0.433510
```

Symptoms of multi-collinearity

- There are high correlations between Age and Pulse ($r = 0.619$), Weight and BSA ($r = 0.875$), Weight and Pulse ($r = 0.659$).
- By deleting Age, it causes dramatic changes in the estimate of BSA, Dur and Pulse (even there is a change of sign).
- By deleting Pulse, it causes dramatic changes in the estimate of BSA.
- The above give evidence for the existence of multi-collinearity.

Variance inflation factors

- To assess multicollinearity for the jth predictor, regress $\boldsymbol{x}_j$ on the remaining predictors X_{-j}.
- Let R_j^2 be the coefficient of multiple determination from this auxiliary regression. Then a larger R_j^2 indicates a higher multicollinearity
- We examine R_j^2. Note that

$$R_j^2 = \frac{SSR_j}{SST_j} = 1 - \frac{SSE_j}{SST_j} = 1 - \frac{\boldsymbol{x}_j^\top (I - H_{-j})\boldsymbol{x}_j}{\sum_{i=1}^n (x_{ij} - \bar{x}_j)^2}$$

- Now recall the original regression. We have

$$\mathrm{Var}(\hat{\boldsymbol{\beta}}_j) = \frac{\sigma^2}{\boldsymbol{x}_j^\top (I - H_{-j})\boldsymbol{x}_j} = \frac{\sigma^2}{\mathrm{SST}_j(1 - R_j^2)} = \frac{\sigma^2}{\mathrm{SST}_j} \cdot \frac{1}{1 - R_j^2}.$$

  Therefore, $1/(1 - R_j^2)$ can be seen an inflation factor.
- We define the variance inflation factor VIF_j : $\mathrm{VIF}_j = \frac{1}{1-R_j^2}$. A larger VIF_j indicates more severe multicollinearity for $\boldsymbol{x}_j$.

An R function for the computation of VIF

- The function `vifChen` can be used to compute VIF which is defined as

```
vifChen=function(object) {
## Input:
# object -- a fitted lm object. The argument x of lm
            must be set as x=TRUE.
## Output:
# vif -- variance inflation factors.
  X = object$x[,-1]
  V = vcov(object)[-1,-1]
  n = dim(X)[1]
  sigma=summary(object)$sigma
  v = diag(V)
  S = diag(var(X))*(n-1)
  vif = v*S/sigma^2
  vif
}
```

- Usage: `vifChen(object)`

Detecting multicollinearity by VIF

- Values of VIF up to 5 are usually considered uncritical, values above 5 indicate a considerable multicollinearity.
- To remedy for the multicollinearity, we can remove the predictors with the largest VIF one at a time until all VIF reduce below 5.
- For the blood pressure data, the VIFs are computed as

```
> fit.full=lm(BP~Age+Weight+BSA+Dur+Pulse+Stress,x=TRUE,
              data=bp.dat)
> vifChen(fit.full)
       Age   Weight      BSA      Dur    Pulse   Stress
  1.762807 8.417035 5.328751 1.237309 4.413575 1.834845
```

- `Weight` has the largest VIF. Remove `Weight` and obtain the following.

```
> regB = lm(BP~Age+BSA+Dur+Pulse+Stress, x=TRUE, data=bp.dat)
> vifChen(regB)
       Age      BSA      Dur    Pulse   Stress
  1.703115 1.428349 1.237151 2.360939 1.502936
```

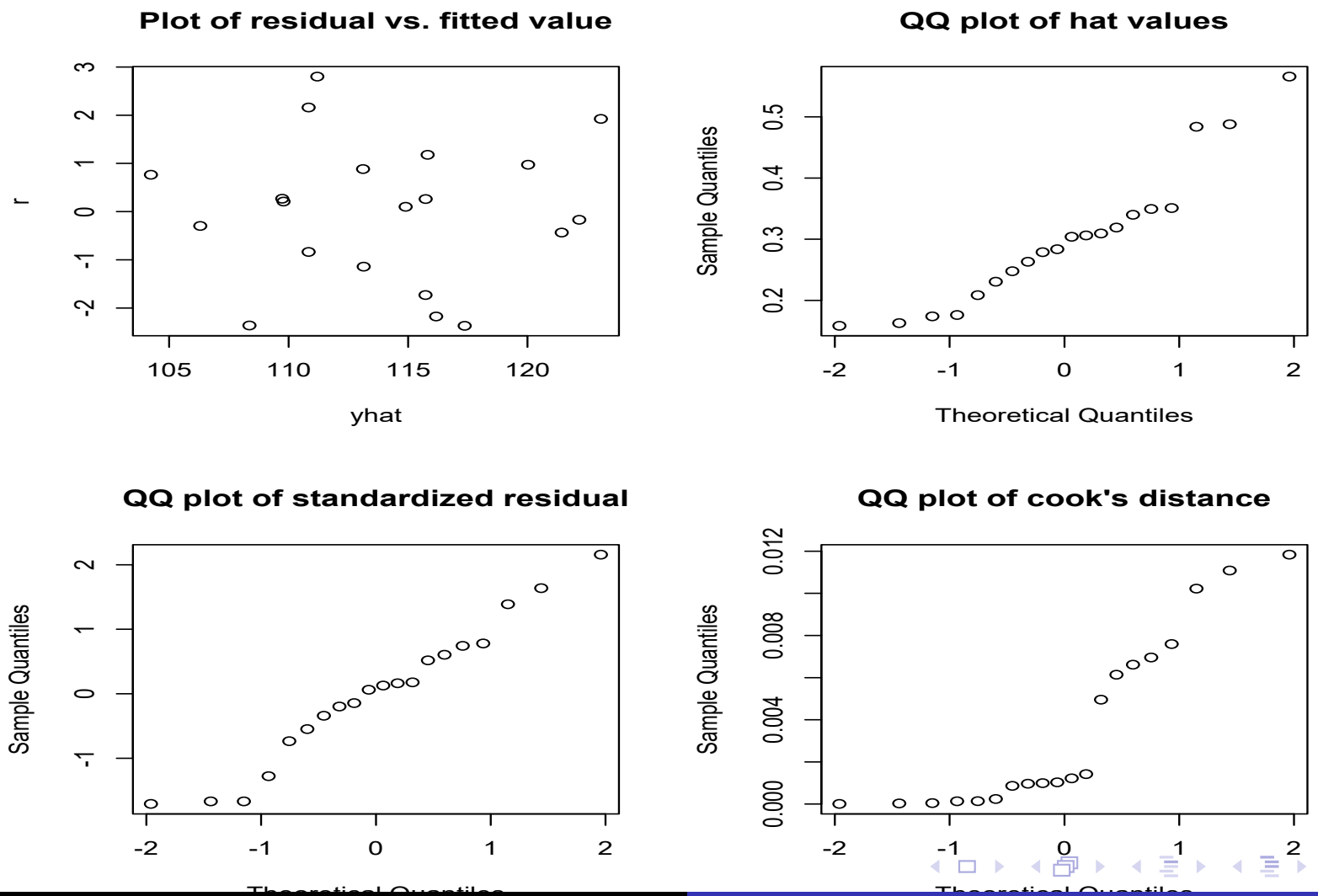
- Removing `Weight` from the model greatly reduces the multicollinearity.

Remedy 1: variable removal based on VIF

- After removing Weight (which has the highest VIF 8.417), the largest VIF among the remaining predictors is 2.36, the collinearity is no longer serious. The data analysis can be carried out with the other predictors.
- The forward selection procedure produces the model including all the other predictors but `Dur`. The fitted results are

```
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  5.50082    8.96912   0.613  0.54886
BSA         24.48547    3.34670   7.316 2.54e-06 ***
Pulse        0.46830    0.15156   3.090  0.00747 **
Age          0.57301    0.19805   2.893  0.01115 *
Stress      -0.01621    0.01258  -1.289  0.21704
```

Diagnostics on refitted model



BP example: conclusion

- The model

$$\text{BP}=\text{Age}+\text{BSA}+\text{Pulse}+\text{Stress}$$

  is an adequate model.
- The three predictors, `BSA`, `Pulse`, `Age`, have significant effects on the blood pressure. Specifically, the increase of any of these predictor values causes an increase in blood pressure.
- Multicollinearity problem:
  - Weight is removed because it has the largest VIF
  - After removing Weight, the remaining VIF values indicate that multicollinearity is no longer serious. The reduced model is suitable for further diagnostics and interpretation.
  - If desired, an additional variable selection step may be applied, but it is not necessary here.

Remedy 2: Ridge regression

- Ridge regression is another approach to handle multicollinearity.
- The immediate consequence of multicollinearity is that it renders the matrix $X^\top X$ nearly singular.
- A simple way to remedy the problem is adding a diagonal matrix to $X^\top X$, i.e. $X^\top X + \lambda I$. The ridge regression estimator is then given by

$$\hat{\boldsymbol{\beta}}_{\mathrm{RIG}} = (X^\top X + \lambda I)^{-1} X^\top \boldsymbol{y}.$$

  where $\lambda > 0$ is a parameter to be chosen.
- The ridge regression estimator is in fact the minimizer of the penalized sum of squares given below:

$$\begin{aligned} & \sum_{i=1}^{n} (y_i - \boldsymbol{\beta}_0 - \sum_{j=1}^{p} x_{ij} \boldsymbol{\beta}_j)^2 + \lambda \sum_{j=1}^{p} \boldsymbol{\beta}_j^2 \\ = \quad & \|\boldsymbol{y} - X\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|^2. \end{aligned}$$

Bias of ridge estimates

- The mean of $\hat{\boldsymbol{\beta}}_{\mathrm{RIG}}$ is given by
$$E\hat{\boldsymbol{\beta}}_{\mathrm{RIG}} = (X^\top X + \lambda I)^{-1} X^\top X \boldsymbol{\beta} = \boldsymbol{\beta} - \lambda (X'X + \lambda I)^{-1} \boldsymbol{\beta}.$$
- Thus $\hat{\boldsymbol{\beta}}_{\mathrm{RIG}}$ is biased. The bias is
$$\boldsymbol{\beta} - E\hat{\boldsymbol{\beta}}_{\mathrm{RIG}} = \lambda (X^\top X + \lambda I)^{-1} \boldsymbol{\beta}.$$
- Let Q be the orthogonal matrix such that $X^\top X = Q \Delta Q^\top$, where $\Delta = \mathrm{Diag}(\tau_0, \tau_1, \ldots, \tau_p)$. Thus
$$\boldsymbol{\beta} - E\hat{\boldsymbol{\beta}}_{\mathrm{RIG}} = Q \begin{pmatrix} \frac{\lambda}{\tau_0+\lambda} & 0 & \cdots & 0 \\ 0 & \frac{\lambda}{\tau_1+\lambda} & \cdots & 0 \\ \cdots & \cdots & \cdots & \cdots \\ 0 & 0 & \cdots & \frac{\lambda}{\tau_p+\lambda} \end{pmatrix} Q^\top \boldsymbol{\beta}.$$
- Let $\tilde{\boldsymbol{\beta}}_j$ denote the jth component of $Q^\top \boldsymbol{\beta}$. We have
$$\|\boldsymbol{\beta} - E\hat{\boldsymbol{\beta}}_{\mathrm{RIG}}\|^2 = \sum_{j=0}^{p} \left(\frac{\lambda}{\tau_j + \lambda} \right)^2 \tilde{\boldsymbol{\beta}}_j^2, \quad \tilde{\boldsymbol{\beta}} = Q^\top \boldsymbol{\beta}$$
- The bias increases as λ increases.

Variance of ridge estimates

- The variance matrix of $\hat{\beta}_{\mathrm{RIG}}$ is given by

$$\begin{aligned}\mathrm{Var}(\hat{\beta}_{\mathrm{RIG}}) &= \sigma^2(X^\top X+\lambda I)^{-1}X^\top X(X^\top X+\lambda I)^{-1}\\ &= \sigma^2 Q\begin{pmatrix}\frac{\tau_0}{(\tau_0+\lambda)^2} & 0 & \cdots & 0\\ 0 & \frac{\tau_1}{(\tau_1+\lambda)^2} & \cdots & 0\\ \cdots & \cdots & \cdots & \cdots\\ 0 & 0 & \cdots & \frac{\tau_p}{(\tau_p+\lambda)^2}\end{pmatrix}Q^\top.\end{aligned}$$

Thus,

$$\mathrm{TR}(\mathrm{Var}(\hat{\beta}_{\mathrm{RIG}}))=\sigma^2\sum_{j=0}^{p}\frac{\tau_j}{(\tau_j+\lambda)^2}.$$

- The variances decrease as λ increases.

Selection of λ

- Bias-Variance trade-off again. We consider MSE or SPSE.
- The MSE of $\boldsymbol{\beta}$ is given by
$$\text{MSE}(\boldsymbol{\beta}) = \|\boldsymbol{\beta} - E\hat{\boldsymbol{\beta}}_{\text{RIG}}\|^2 + \text{TR}(\text{Var}(\hat{\boldsymbol{\beta}}_{\text{RIG}})).$$
A balance must be striken between the bias and the variance by minimizing MSE.
- In practice, we can select λ by cross-validation (CV).
- Let $\hat{\boldsymbol{\beta}}_{-i}(\lambda)$ be the ridge regression estimate of $\boldsymbol{\beta}$ with parameter λ by deleting the ith observation. The CV score is given by
$$CV(\lambda) = n^{-1}\sum_{i=1}^{n}[y_i - \boldsymbol{x}_{(i)}^\top \boldsymbol{\beta}_{-i}(\lambda)]^2,$$
where $\boldsymbol{x}_{(i)}^\top$ is the ith row vector of the design matrix X.
- The best λ is the minimizer of $CV(\lambda)$.

R function lm.ridge in package MASS

- Usage of `lm.ridge`

```
lm.ridge(formula, data, lambda, ...)

Major arguments:
  formula--expression for the regression model;
  data--a data frame used for the model;
  lambda--A scalar or vector of ridge constants.
Major outputs:
  lambda--vector of lambda values;
  GCV--vector of GCV values.
```

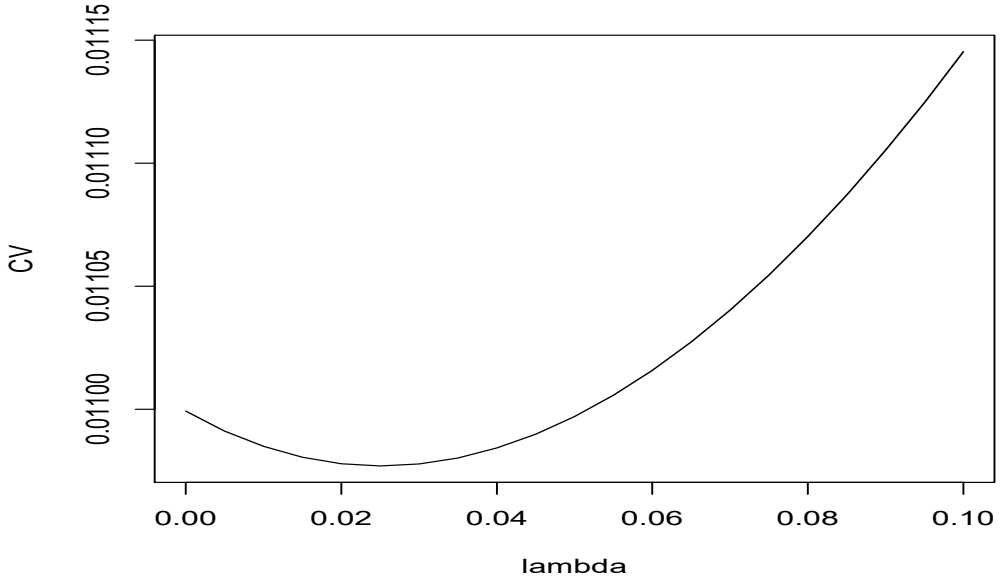
- To get the best λ value, specify the vector λ in `lm.ridge`, from the output, get `GCV` and then identify the λ value corresponding to the smallest GCV value.
- Fit the final model using the best λ value.

BP example: Ridge regression

```
library(MASS)
regRidge = lm.ridge(BP~Age+Weight+BSA+Dur+Pulse+Stress,
                    data=bp.dat,lambda=seq(0,0.1,0.005))
CV=regRidge$GCV
lambda=regRidge$lambda
plot(lambda,CV,type="l")
lambda.best = lambda[order(CV)[1]]
ridge.fit = lm.ridge(BP~Age+Weight+BSA+Dur+Pulse+Stress,
                     data=bp.dat,lambda=lambda.best)
ridge.fit
```

Estimated parameters and CV plot

```
             Age Weight    BSA    Dur   Pulse Stress
-12.6490 0.7003 0.9599 3.9850 0.0685 -0.0780 0.0053
```



Remark. $\lambda = 0$ doesn't yield the lowest risk. It is common in multivariate LRM.

Remarks on ridge regression

- In the BP example, ridge regression gives more stable coefficient estimates in the presence of multicollinearity.
- Compared with LSE, the ridge estimates are shrunk toward zero and are less variable.
- The tuning parameter λ controls the amount of shrinkage:

$$\lambda = 0 \;\Rightarrow\; \text{LSE estimates}, \qquad \text{larger } \lambda \;\Rightarrow\; \text{stronger shrinkage.}$$

- Ridge regression is preferred when multicollinearity is severe and the main goal is stable estimation or prediction.
- It is less suitable when the main goal is variable selection or classical inference on individual regression coefficients.

Remedy 3: PCR for multicollinearity

- Principal components regression (PCR) is another approach to handle multicollinearity.
- Let X be the centered design matrix and the eigen-decomposition
$$X^\top X = Q\Delta Q^\top,$$
where $Q = (q_1, \ldots, q_p)$ is an orthogonal matrix and $\Delta = \mathrm{Diag}(\tau_1, \ldots, \tau_p)$ with $\tau_1 \geq \cdots \geq \tau_p \geq 0$.
- The principal components are (project X on Q)
$$Z = XQ = (\boldsymbol{z}_1, \ldots, \boldsymbol{z}_p).$$
- PCR fits the regression of y on the first k principal components:
$$y = \alpha_0 + \alpha_1 z_1 + \cdots + \alpha_k z_k + \varepsilon.$$
- Thus, the PCR estimator using the first k components is
$$\hat{\boldsymbol{\beta}}_{\mathrm{PCR}} = Q_k \hat{\boldsymbol{\alpha}} = \sum_{j=1}^{k} \frac{q_j q_j^\top \boldsymbol{y}}{\tau_j}.$$

Connection between PCR and LSE

- In terms of q_j, the LSE can be rewritten as

$$\hat{\boldsymbol{\beta}}_{\mathrm{LSE}} = \sum_{j=1}^{p} \frac{q_j q_j^\top \boldsymbol{y}}{\tau_j}.$$

  In other words, it is PCR estimate when $k = p$.
- It also concludes the effect of multicollinearity. Under multicollinearity, $\tau_p \approx 0$, which makes $q_p q_p^\top \boldsymbol{y}$ extremely important in the estimation. Any changes in $\boldsymbol{y}$ in this direction causes a big change in $\hat{\boldsymbol{\beta}}$
- PCR removes those with smallest τ_j, so that the result is stable

Bias of PCR estimates

- The mean of $\hat{\boldsymbol{\beta}}_{pcr}$ is

$$E(\hat{\boldsymbol{\beta}}_{pcr}) = Q_k \Delta_k^{-1} Q_k^\top X^\top X \boldsymbol{\beta} = Q_k Q_k^\top \boldsymbol{\beta}.$$

- Thus $\hat{\boldsymbol{\beta}}_{pcr}$ is biased when $k < p$. The bias is

$$\boldsymbol{\beta} - E(\hat{\boldsymbol{\beta}}_{pcr}) = (I - Q_k Q_k^\top)\boldsymbol{\beta}.$$

- Let $\tilde{\boldsymbol{\beta}}_j$ denote the jth component of $Q^\top \boldsymbol{\beta}$. Then $\boldsymbol{\beta} - E(\hat{\boldsymbol{\beta}}_{pcr}) = \sum_{j=k+1}^{p} \tilde{\boldsymbol{\beta}}_j q_j$, and hence

$$\|\boldsymbol{\beta} - E(\hat{\boldsymbol{\beta}}_{pcr})\|^2 = \sum_{j=k+1}^{p} \tilde{\boldsymbol{\beta}}_j^2.$$

- The bias decreases as more principal components are included.

Variance of PCR estimates

- The variance matrix of $\hat{\beta}_{pcr}$ is given by $\mathrm{Var}(\hat{\beta}_{pcr}) = \sigma^2 Q_k \Delta_k^{-1} Q_k^\top$.
- Equivalently,

$$\mathrm{Var}(\hat{\beta}_{pcr}) = \sigma^2 Q \begin{pmatrix} \tau_1^{-1} & 0 & \cdots & 0 & \cdots & 0 \\ 0 & \tau_2^{-1} & \cdots & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & & \vdots \\ 0 & 0 & \cdots & \tau_k^{-1} & \cdots & 0 \\ \vdots & \vdots & & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & \cdots & 0 \end{pmatrix} Q^\top.$$

- Thus

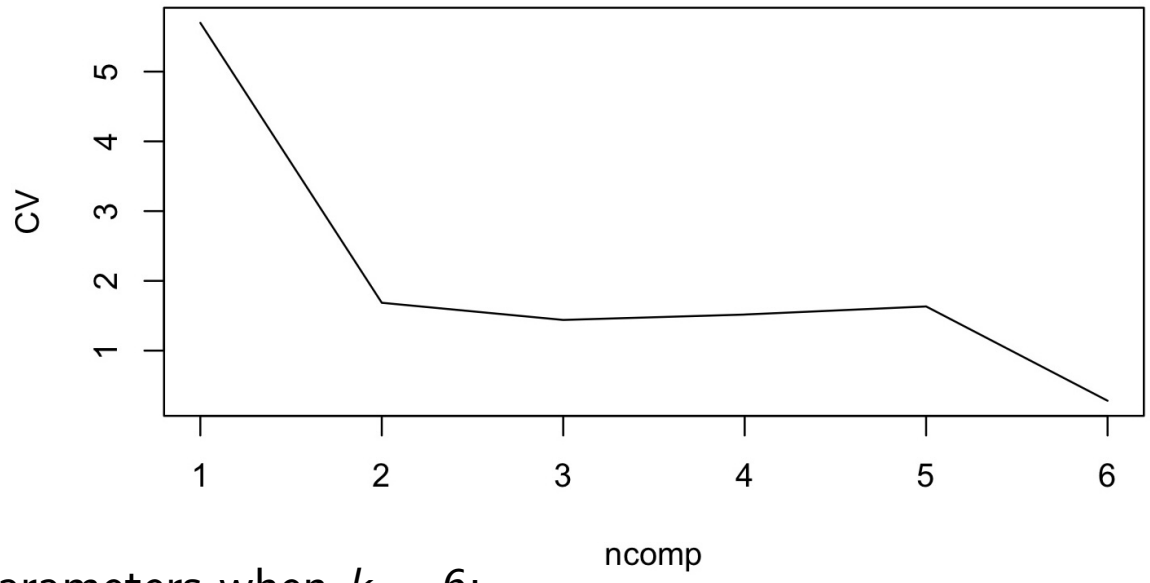
$$tr\{\mathrm{Var}(\hat{\beta}_{pcr})\} = \sigma^2 \sum_{j=1}^{k} \frac{1}{\tau_j}.$$

- The variances increase as more principal components are included.

BP example: Principal components regression

```
library(pls)
regPCR = pcr(BP~Age+Weight+BSA+Dur+Pulse+Stress,
             data=bp.dat, scale=TRUE, validation="CV")
CV = MSEP(regPCR)$val[1,1,-1]
ncomp = 1:length(CV)
plot(ncomp,CV,type="l")
# validationplot(regPCR, val.type="MSEP")
k.best = ncomp[order(CV)[1]]
pcr.fit = pcr(BP~Age+Weight+BSA+Dur+Pulse+Stress,
              data=bp.dat, scale=TRUE, ncomp=k.best)
coef(pcr.fit, ncomp=k.best)
```

Estimated parameters and CV plot



Estimated parameters when $k = 6$:

Variable	Age	Weight	BSA	Dur	Pulse	Stress
Estimate	1.7585	4.1657	0.5154	0.1467	−0.3213	0.2066

Remark. In this example, the CV criterion selects $k = 6$ principal components.

Thus PCR is essentially the same as the least squares fit.

Remarks on principal components regression

- PCR reduces multicollinearity by regressing on a few orthogonal principal components.
- The number of principal components k controls the amount of dimension reduction:

  small $k \Rightarrow$ larger bias but smaller variance,

  large $k \Rightarrow$ smaller bias but larger variance.
- PCR performs hard truncation, while ridge regression performs continuous shrinkage.
- In the BP example, the CV criterion selects all 6 principal components. So PCR is the same as the least squares fit in this case.

Remedies for non-normality

- Non-normality is often accompanied by non-constant variance.
  - For normal dist, the mean and variance are separate parameters, i.e., the variance is a constant when the fitted value changes
  - The violation of normality usually goes together with the violation of constancy of variance.
- A transformation of the response may improve both normality and variance stability.
- In some cases, the choice of transformation is suggested by the nature of the response.
- In other cases, a more systematic choice of transformation is needed.

Variable transformation as a remedy for non-normality

- A variance stabilization transformation can help to rectify simultaneously the discrepancy in both normality and constancy of variance.
- If the relationship between the variance and the mean is known, the desired variance stabilization transformation can be derived from that relationship.
- In general, the transformation can be explored by Box-Cox transformation.

Variance-stabilizing transformation

- Consider non-constant variance. Suppose

$$\mathrm{Var}(Y_i \mid \boldsymbol{x}_i) = \sigma^2(\mu_i), \qquad \mu_i = E(Y_i \mid \boldsymbol{x}_i).$$

- We want a transformation $g(Y)$ such that

$$\mathrm{Var}\{g(Y_i) \mid \boldsymbol{x}_i\} \text{ is approximately constant.}$$

- A first-order approximation gives

$$\mathrm{Var}\{g(Y_i) \mid \boldsymbol{x}_i\} \approx [g'(\mu_i)]^2 \sigma^2(\mu_i).$$

- So we choose g such that

$$g'(\mu) \propto \frac{1}{\sigma(\mu)}.$$

- It is called the Variance Stabilizing Transformation (VST).

Senile dementia patients example: Proportion data

A trial was conducted to compare two methods for the training of senile dementia patients.

- Two groups of patients with sizes 11 and 8 are recruited.
- The two groups are trained with different methods.
- After the completion of the training, patients are asked to take 20 tests involving activities of daily living such as unlocking a door, tying one's shoe laces, etc.
- The response from each patient is the proportion of his or her tests that are successful.

SDP example: the data

	Group 1		Group 2	
	Y (original)	A (transformed)	Y (original)	A (transformed)
	0.05	0.226	0	0
	0.15	0.398	0.15	0.398
	0.35	0.633	0	0
	0.25	0.524	0.05	0.226
	0.20	0.464	0	0
	0.05	0.226	0	0
	0.10	0.322	0.05	0.226
	0.05	0.226	0.10	0.322
	0.30	0.580		
	0.05	0.226		
	0.25	0.524		
Mean	0.164	0.395	0.044	0.147
sd	0.112	0.158	0.056	0.166

Variance assumption for proportions

- Proportion response has similar natures to a binomial random variable $Bin(n, p)$:
  - (i) The variance $np(1-p)$ depends on the mean np;
  - (ii) The variance is smaller if the mean is closer to either zero or 1, larger if the mean is closer to 0.5.
- In the example, the mean of Group 2 is closer to zero than that of Group 1, and the variance of Group 2 ($0.056^2 = 0.003136$) is about a quarter of that of Group 1 ($0.112^2 = 0.012544$).
- The variance of proportion response can be approximated by $\sigma^2(\mu) = c\mu(1-\mu)$. Thus the VST is given by

$$g(Y) = \arcsin\sqrt{Y}.$$

SDP example: R codes

```
y=c(0.05,0.15,0.35,0.25,0.2,0.05,0.1,0.05,0.3,0.05,0.25,
0,0.15,0,0.05,0,0,0.05,0.1)
g=factor(c(rep(1,11),rep(2,8)))
org.fit=lm(y~g); r.o=org.fit$resid; y.o = org.fit$fitted

y.t=asin(sqrt(y))
tra.fit=lm(y.t~g); r.t=tra.fit$resid;y.t = tra.fit$fitted

par(mfrow=c(2,1))
plot(y.o,r.o,main="Residual plot of original fitting")
plot(y.t,r.t,main="Residual plot of transformed fitting")

summary(org.fit);summary(tra.fit)
```

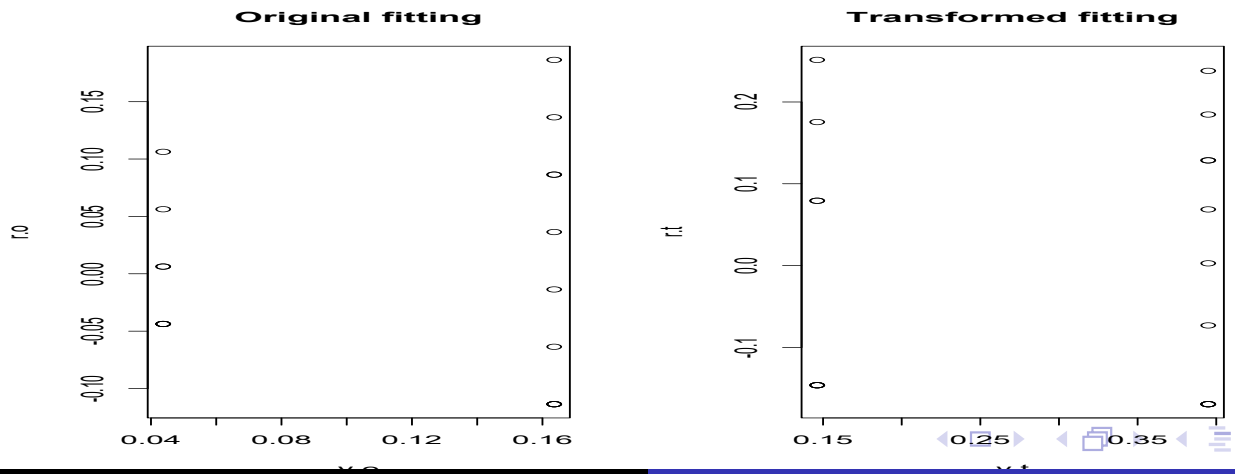
SDP example: Results

- Fitted results:

```
Original:    Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.1636     0.0281   5.824 2.03e-05
g2           -0.1199     0.0433  -2.769   0.0131

Transformed: Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.39500    0.04856   8.134 2.91e-07
g2           -0.24869    0.07484  -3.323  0.00402
```

- Residual plot:



SDP example: analysis

- The model based on the original data is not appropriate as indicated by the difference of sd in the two groups and also shown in the residual plot.
- The residual plot alone does not give a clue on the nature of the variance, the VST is provided by the nature of the data.
- By the p-value 0.00402 of the t-value -3.323, it can be drawn that the two training methods are significantly different. The first method is better.
- The analysis can also be carried out by a two sample t-test with the test statistic

$$T = \frac{\bar{Y}_1 - \bar{Y}_2}{s}\sqrt{\frac{n_1 n_2}{n_1 + n_2}},$$

where

$$s = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 1}}.$$

Under the normality assumption and the null hypothesis of zero difference, $T \sim t_{17}$.

Trial of Vaccination (ToV) example: Count data

In many clinical trials, the response variable is a count, e.g., number of microrganisms, number of attacks of angina pectoris, etc. The following trial provides an example.

Trial of vaccination

- Purpose: to assess the effect of vaccination on heat-killed bacilli.
- Case group ($n_1 = 7$): vaccinated.
- Control group ($n_2 = 6$): not vaccinated.
- Measurements: number of oral lactobacilli in the saliva.

ToV example: data

	Case		Control	
	Y	$\sqrt{Y}$	Y	$\sqrt{Y}$
	7,925	89.02	3,158	56.20
	15,643	125.07	3,669	60.57
	17,462	132.14	5,930	77.01
	10,805	103.95	5,697	75.48
	9,300	96.44	8,331	91.27
	7,538	86.82	11,822	108.73
	6,297	79.35		
Mean	10,710.0	101.827	6,434.5	78.21
sd	4,266.4	19.946	3,218.8	19.527

The nature of count response

- A count response variable can usually be approximated by a Poisson random variable.
- For Poisson distribution, $\sigma^2(\mu) = \mu$. In general, for count response variables, the variance is proportional to the mean, i.e., $\sigma^2(\mu) = c\mu$.
- The VST of count response Y is given by

$$g(Y) = \sqrt{Y}. \quad (g'(Y) = 1/\sqrt{Y} = 1/\sigma(Y))$$

- In the data table on the previous slide, the variances of the two groups are quite different, but

$$\frac{s_1}{\sqrt{\bar{Y}_1}} = 41.2 \approx 40.1 = \frac{s_2}{\sqrt{\bar{Y}_2}},$$

which suggests that the variance is proportional to the mean.

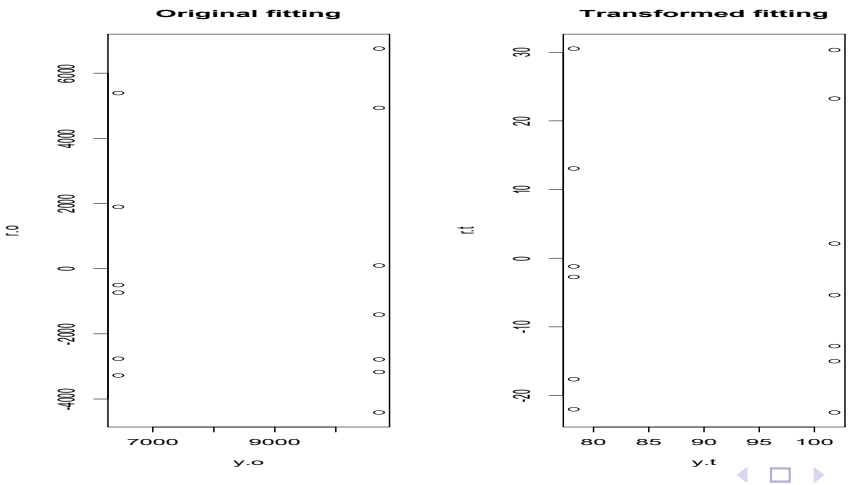
ToV example: Results

- Fitted results:

```
Original      Estimate Std. Error t value Pr(>|t|)
(Intercept)     10710       1446   7.406 1.35e-05
T2              -4276       2129  -2.009   0.0698

Transformed: Estimate Std. Error t value Pr(>|t|)
(Intercept)   101.828      7.468  13.636  3.1e-08
T2            -23.619     10.992  -2.149   0.0548
```

- Residual plot:



ToV example: analysis

- The model based on the original data does not have the constancy of variance but the transformed model does, as shown in the residual plots.
- The transformation is suggested by the the ratio of the sample standard deviation and the square root of the sample mean, as well as the general nature of count data.
- By the p-value 0.0548 of the t-value -2.149, it can be drawn that the case and control group are significantly different at a level slightly higher than 0.05, i.e., the vaccination does have a significant effect. The vaccination can reduce the number of the oral lactobacilli in the saliva by an estimated expected value about 24.

Miller Lumber Company (MLC) example: Continuous data

- In a survey to investigate how certain factors affect the number of customers of the Miller lumber Company, the following data are collected from 110 census tracts:

  y–number of customers who visited the store from the tract,
  x_1–number of housing units,
  x_2–average income,
  x_3–average housing unit age,
  x_4–distance to nearest competitor,
  x_5–distance to store.

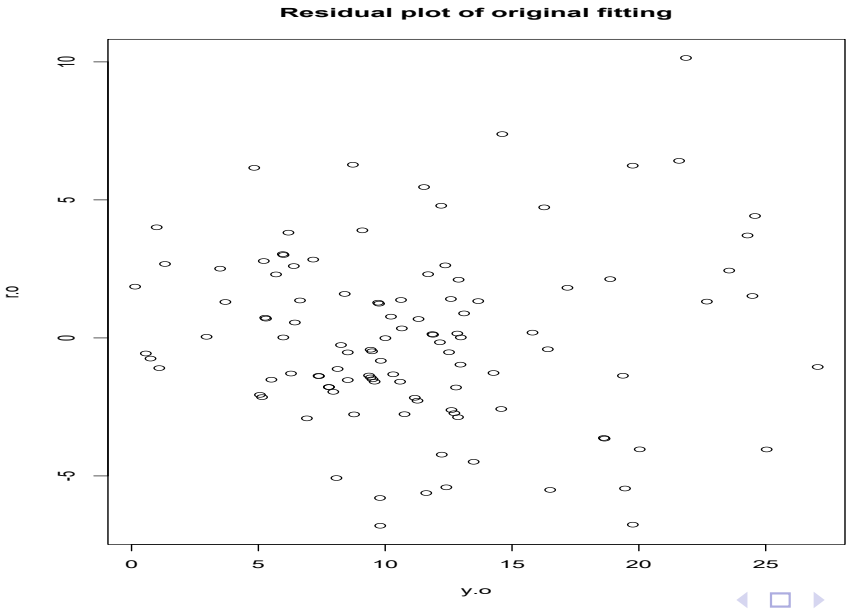
- Data is provided in the file `MLC.txt` on Canvas.

MLC example: Fitting of original y

- R codes:

```
mlc.dat = read.table("D://Rsession/MLC.txt",header =TRUE)
y=mlc.dat$NoC; x1=mlc.dat$HU; x2=mlc.dat$AvI;
x3=mlc.dat$AvA; x4=mlc.dat$CD; x5=mlc.dat$SD
orig.fit = glm(y~x1+x2+x3+x4+x5,data=wk.dat1)
```

- Diagnostic plot:

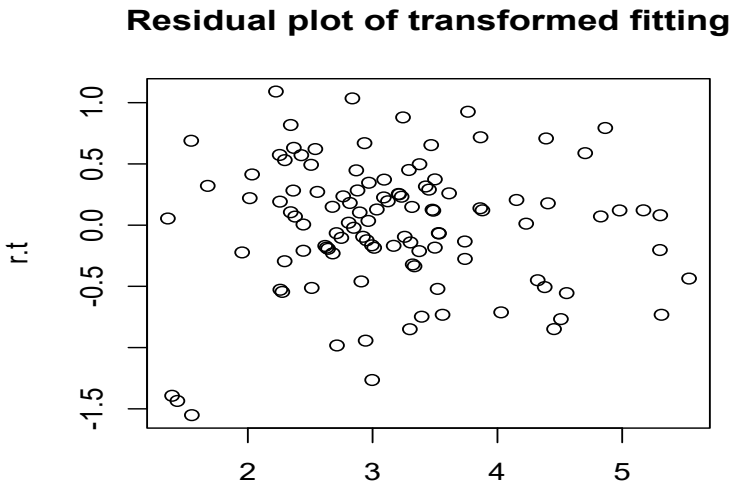


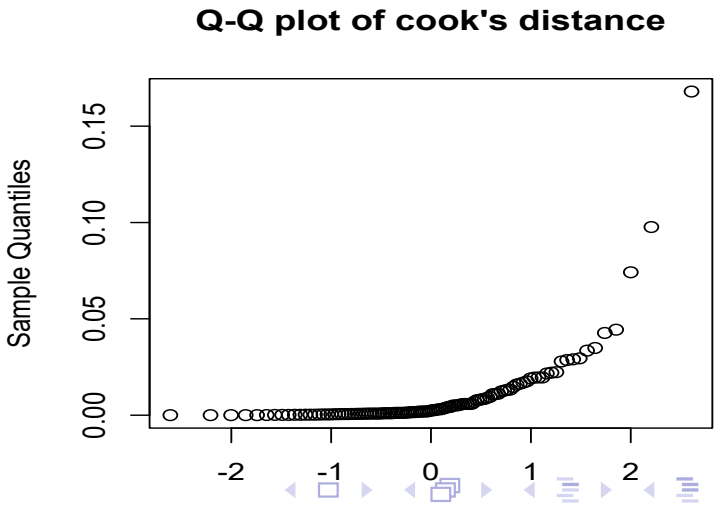
MLC example: Fitting of transformed $yy = \sqrt{y}$

- The residual plot of the original fitting indicates non-constancy of variance. By the nature of the data, a square root transformation is suggested:

```
yy=sqrt(y); wk.dat1=data.frame(yy,x1,x2,x3,x4,x5)
tran.fit = glm(yy~x1+x2+x3+x4+x5,data=wk.dat1)
```

- Diagnostic plots:



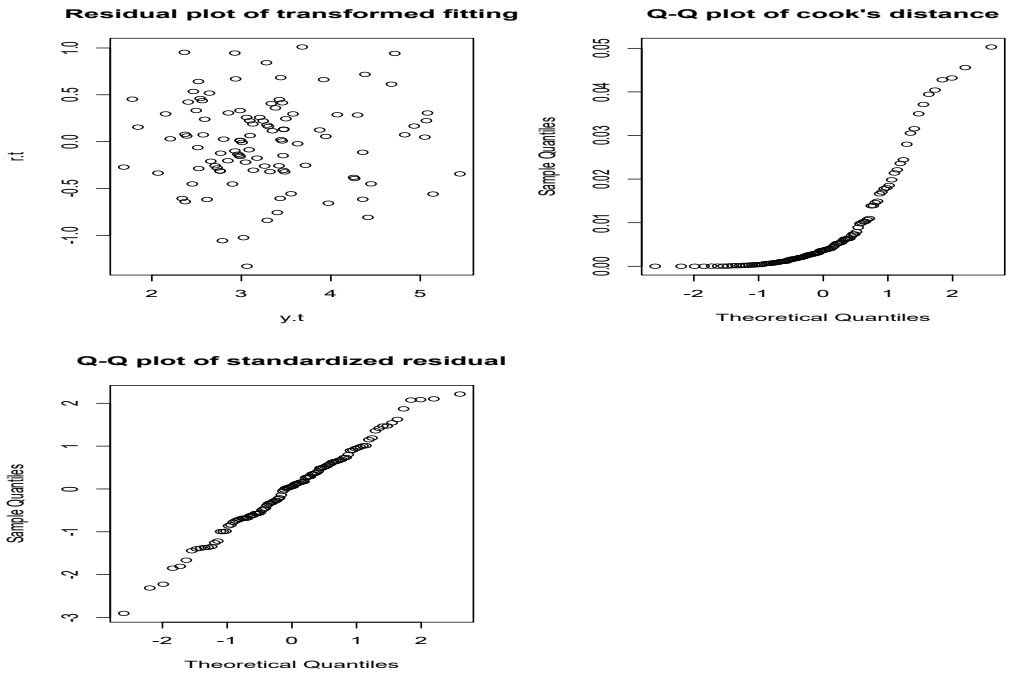


MLC example: Transformed fitting deleting outliers

- R codes:

```
wk.dat2=wk.dat1[-c(7,37,45),];
tran.fit2 = glm(yy~x1+x2+x3+x4+x5,data=wk.dat2)
```

- Diagnostic plots:



MLC exmaple: discussion

- The diagnostics for the model fitted to the original response y reveals that its variance is dependent of its mean.
- The transformation $\sqrt{y}$ according to the nature of the count data is made.
- The model fitted to the transformed response gets rid of the non-constancy of variance. A few influential outliers show up.
- After deleting the influential outliers, the model becomes adequate without obvious discrepancies from normal model assumptions.
- The following variance inflation factor values do not reveal any serious multicollinearity:

```
x1       x2       x3       x4       x5
1.710487 2.019595 1.027386 1.550676 1.652166
```

MLC example: final result and inference

```
Estimate Std. Error t value Pr(>|t|)
(Intercept)  4.283e+00  3.153e-01  13.584  < 2e-16
x1           9.011e-04  2.266e-04   3.976 0.000132
x2          -1.637e-05  3.481e-06  -4.702 8.17e-06
x3          -5.469e-03  2.739e-03  -1.996 0.048585
x4           2.832e-01  3.793e-02   7.466 2.99e-11
x5          -2.279e-01  2.546e-02  -8.951 1.83e-14
```

- All the five predictor variables have significant effects on the response varianble.
- The larger the number of housing units x_1, the further away from nearest competitor (x_4), the survey tract has more customers visiting the MLC stores.
- On the contrast, the average income (x_2), the average housing unit age (x_3) and the distance to MLC stores have a negative effect on the number of customers visiting MLC stores.

Choosing a transformation

- For proportion data, a variance-stabilizing transformation is often available.
- For count data, simple transformations such as $\sqrt{y}$ are often useful.
- For continuous responses, the appropriate transformation is usually less obvious.
- We next consider the Box-Cox approach for choosing a suitable transformation.

Box-Cox transformation

- The Box-Cox transformation provides a tool to determine the transformation. It is given by

$$g(Y) = \frac{Y^\lambda - 1}{\lambda}.$$

  The parameter λ can be determined by the data.
- When $\lambda = 0$, the Box-Cox transformation is given by

$$g(Y) = \log(Y), \text{ since } \lim_{\lambda \to 0} \frac{Y^\lambda - 1}{\lambda} = \log(Y).$$

- The value of λ needs to be determined for particular problems.

λ determined from the power variance function

Let $\sigma_i \propto \mu_i^{\alpha}$, where σ_i and μ_i are, respectively, the standard deviation and the mean of the ith treatment effect. The λ in Box-Cox transformation is determined by the VST.

$\sigma_i \propto \mu_i^{\alpha}$	α	$\lambda = 1 - \alpha$	Transformation
$\sigma_i \propto \mu_i^{3}$	3	-2	reciprocal squared
$\sigma_i \propto \mu_i^{2}$	2	-1	reciprocal
$\sigma_i \propto \mu_i^{3/2}$	3/2	-1/2	reciprocal square root
$\sigma_i \propto \mu_i$	1	0	log
$\sigma_i \propto \mu_i^{1/2}$	1/2	1/2	square root
$\sigma_i \propto$ constant	0	1	no transformation
$\sigma_i \propto \mu_i^{-1/2}$	-1/2	3/2	3/2 power
$\sigma_i \propto \mu_i^{-1}$	-1	2	square

Determination of α

- *Method I:* If observations are grouped, for each group, compute s_i and $\bar{y}_{i\cdot}$. Fit the regression model

$$\log s_i = \beta_0 + \beta_1 \log \bar{y}_{i\cdot} + e_i.$$

  If observations are not grouped, replace s_i and $\bar{y}_{i\cdot}$ with $|r_i|$ and $\hat{y}_i$ respectively. Take α as the estimate of β_1.
- *Method II* (only works for grouped observations): Select a few number of α values, say α_k, $k = 1, \ldots, K$. For each k, compute

$$R_k = \frac{\max_i \frac{s_i}{\bar{y}_{i\cdot}^{\alpha_k}}}{\min_i \frac{s_i}{\bar{y}_{i\cdot}^{\alpha_k}}}.$$

  Select the α_k with the smallest R_k.

Direct determination of *lambda*

For grouped observations:

- Select a few number of λ values, say λ_k, $k = 1, \ldots, K$.
- For each k, make the transformation $y_{ij} \to y_{ij}^{\lambda_k}$. With the transfoumed data, compute $s^2_{\lambda_k i}, i = 1, \ldots, g$.
- Select the λ_k such that

$$\frac{\max s^2_{\lambda_k i}}{\min s^2_{\lambda_k i}}$$

  is the closest to 1.

For non-grouped observations:

- Select a few number of λ values, say λ_k, $k = 1, \ldots, K$.
- For each k, make the transformation $y_i \to y_i^{\lambda_k}$. Analyze the regression models with $y_i^{\lambda_k}$ as the response variable.
- Select the λ_k such that the MSE(λ_k) is the smallest.

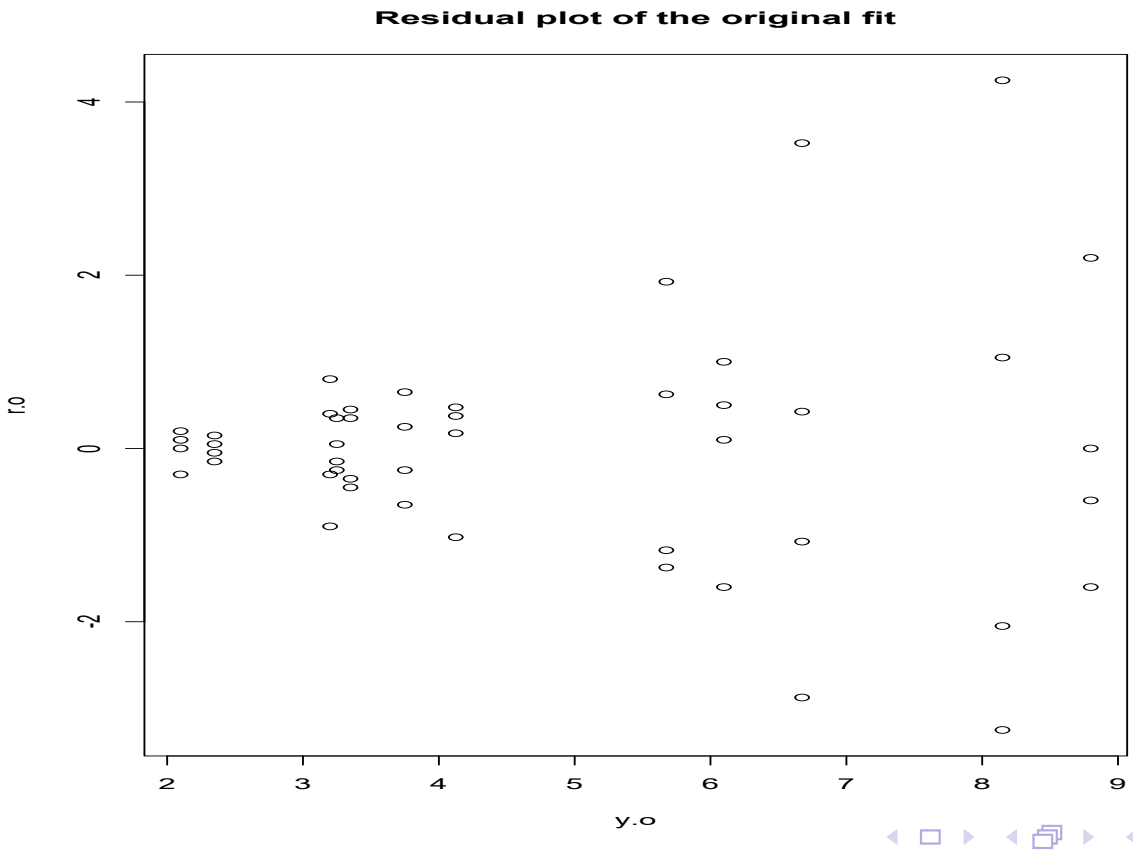
Trial of animal survival (TAS) example

- Purpose of trial: comparison of effects of several combinations of poisons and treatments on the survival of animals.
- Group assignment: combinations are randomly assigned to the animals.
- Measurements: survival time in hours after treatment.

Group (x)	Measurements	$\bar{y}_i$	s_i
1	3.1, 4.5, 4.6, 4.3	4.125	0.695
2	8.2, 11.0, 8.8, 7.2	8.800	1.608
3	4.3, 4.5, 6.3, 7.6	5.675	1.567
4	4.5, 7.1, 6.6, 6.2	6.100	1.128
5	3.6, 2.9, 4.0, 2.3	3.200	0.715
6	9.2, 6.1, 4.9, 12.4	8.150	3.363
7	4.4, 3.5, 3.1, 4.0	3.750	0.569
8	5.6, 10.2, 7.1, 3.8	6.675	2.710
9	2.2, 2.1, 1.8, 2.3	2.100	0.216
10	3.0, 3.7, 3.8, 2.9	3.350	0.465
11	2.3, 2.5, 2.4, 2.2	2.350	0.129
12	3.0, 3.6, 3.1, 3.3	3.250	0.265

TAS example, fitting to the original data

```
fit.o=lm(y~x); r.o=fit.o$resid; y.o=fit.o$fitted
plot(y.o,r.o,main="Residual plot of the original fit" )
```



TAS example, determination of transformation

- Method I: $\log(s)\tilde{}\log(\bar{y})$:

```
Estimate Std. Error t value Pr(>|t|)
(Intercept)  -3.2026     0.4040  -7.927 1.28e-05
log(ybar)     1.9770     0.2633   7.509 2.04e-05
```

  $\log(|r|)\tilde{}\log(\hat{y})$:

```
Estimate Std. Error t value Pr(>|t|)
(Intercept)   -4.894      3.412  -1.435    0.158
log(y.o)       1.906      2.223   0.857    0.396
```

  In both case $\hat{\alpha} \approx 2$, hence $\lambda = -1$.

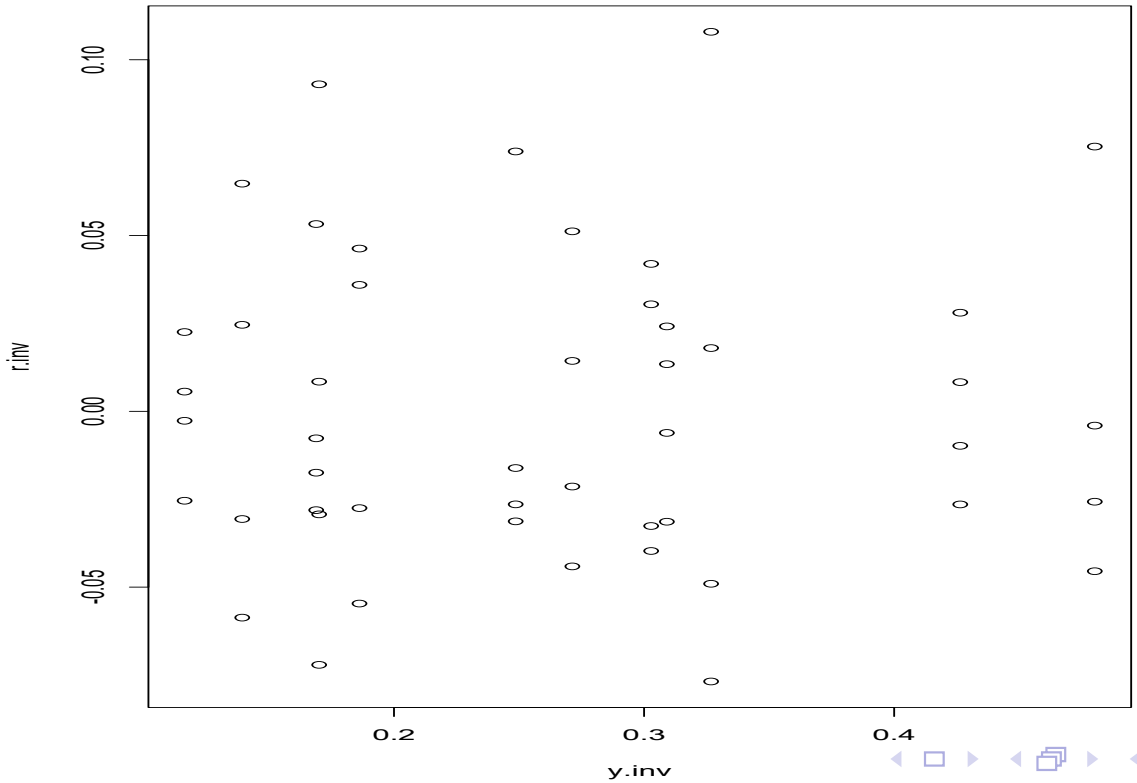
- Method II:

```
alpha=c(-1,-0.5,0.5,1,1.5,2,3); R=NULL
for (i in 1:length(alpha) ) {
    R[i] = max(sd/ybar^alpha[i])/ min(sd/ybar^alpha[i])}
alpha -1.00 -0.50  0.50 1.00 1.50 2.00 3.00
R     90.34 48.51 13.99 7.51 4.38 3.54 9.88
```

  the α value achieving the smallest R_k is 2.

TAS example, inverse transformation

```
yy=1/y;   fit.inv=lm(yy~x)
r.inv=fit.inv$resid;  y.inv=fit.inv$fitted
plot(y.inv,r.inv,main="Residual plot of the transformed fit")
```



TAS example, result and inference

- The fitted results:

```
Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.24869    0.02450  10.151 4.16e-12 ***
x2          -0.13234    0.03465  -3.820 0.000508 ***
x3          -0.06242    0.03465  -1.801 0.080010 .
x4          -0.07972    0.03465  -2.301 0.027297 *
x5           0.07816    0.03465   2.256 0.030252 *
x6          -0.10935    0.03465  -3.156 0.003226 **
x7           0.02270    0.03465   0.655 0.516449
x8          -0.07853    0.03465  -2.267 0.029515 *
x9           0.23158    0.03465   6.684 8.56e-08 ***
x10          0.05421    0.03465   1.565 0.126425
x11          0.17781    0.03465   5.132 1.00e-05 ***
x12          0.06049    0.03465   1.746 0.089349 .
F-statistic: 21.53 on 11 and 36 DF,  p-value: 1.289e-12
```

- The effects of the combinations of poisons and treatments are significantly different (F-statistic: 21.53 with p-value almost zero).
- All the combinations except combination 7 and 10 are significantly different from combination 1, which can be inferenced directly from the above table.
- The full pairwise comparison can be made with additional information on the covariance matrix of the coefficient estimates.

Peptic ulcer (PU) example

Lysozyme levels in the gastric juice are to be compared between patients with peptic ulcer and normal controls. The following are the data for 29 patients and 30 controls.

Group 1 ($n_1 = 29$)			Group 2 ($n_2 = 30$)		
0.2	4.9	17.6	0.2	2.5	8.8
0.3	5.0	18.9	0.3	2.8	9.1
0.4	5.3	20.7	0.4	3.6	10.3
1.1	7.5	24.0	0.7	4.8	15.6
2.0	9.8	25.4	1.2	4.8	16.1
2.1	10.4	40.0	1.5	5.4	16.5
3.3	10.9	42.2	1.5	5.7	16.7
3.8	11.3	50.0	1.9	5.8	20.0
4.5	12.4	60.0	2.0	7.5	20.7
4.8	16.2		2.4	8.7	33.0
$\bar{y}_1 = 14.31, s_1 = 15.74$			$\bar{y}_2 = 7.68, s_2 = 7.85$		

PU example, determination of transformation

- Computation of transformed s_1/s_2:

```
lambda=c(-1,-0.5,0.001,0.5); s1=NULL; s2=NULL
for (i in 1:4) {
z1=(y1^lambda[i]-1)/lambda[i];
z2=(y2^lambda[i]-1)/lambda[i]
s1[i]=sqrt(var(z1)); s2[i]=sqrt(var(z2))
}
s.ratio = s1/s2
lambda  -1.000000 -0.500000 0.001000 0.500000
s.ratio  1.025093  1.035331 1.125875 1.439527
```

- The best transformation among the four is the reciprocal transformation. Its ratio s_1/s_2 is nearest to 1.

PU example, model fitting and inference

- Fitted results:

```
y=c(y1,y2); y.t=1/y;x=factor(c(rep(1,29),rep(2,30)))
fit.o=lm(y~x); fit.t =lm(y.t~x)
summary(fit.o)
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   14.310      2.297   6.229 6.05e-08 ***
x2            -6.627      3.222  -2.057   0.0443 *
summary(fit.t)
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.52713    0.20811   2.533   0.0141 *
x2           0.09955    0.29185   0.341   0.7343
```

- The difference is significant with a p-value 0.0443 based on the original data. But it is not significant based on transformed data.
- An inference is valid only if the assumption of model is satisfied. The original data violates the assumption and hence the inference is invalid. The reference must be based on the transformed data.